

WIRELESS LINK BUDGET ANALYSIS REPORT

Course: 3171608 - Wireless Communication (WC) | Micro-Project

Scenario: 4G LTE Urban Macro Cell (1800 MHz Band 3) | Generated: 2026-10-05 23:29:38 | Model: COST-231 Hata Model

LINK STATUS: X FAIL / MARGIN DEFICIT — LINK DOWN / FAILED (Margin Deficit: 10.74 dB)
Received Power: -110.74 dBm | Sensitivity: -100.00 dBm | Fade / Link Margin: -10.74 dB

Executive Performance Metrics

EIRP	57.00 dBm (501.187 W)	Received Power (Prx)	-110.74 dBm (0.01 pW)
Free Space Path Loss	103.57 dB	Model Path Loss	145.74 dB
Total Environmental Loss	22.00 dB	Total Attenuation	169.74 dB
Thermal Noise Floor	-93.96 dBm	Signal-to-Noise Ratio (SNR)	-16.78 dB
Max Achievable Distance	0.982 km	Shannon Capacity	0.60 Mbps
Max Allowable Path Loss	135.00 dB	Carrier Wavelength (λ)	0.1666 m (16.66 cm)

System Input Parameters

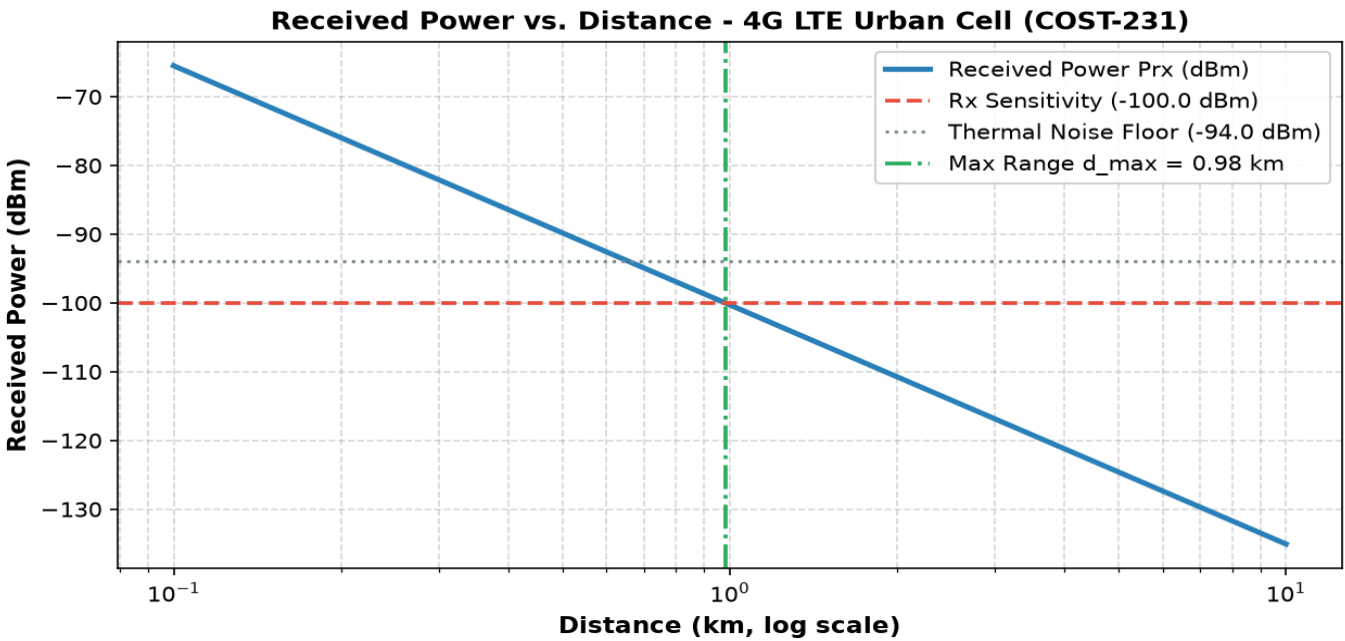
Subsystem	Parameter Name	Value	Units / Notes
Transmitter (Tx)	Tx Output Power (Ptx)	43.00	dBm
	Tx Antenna Gain (Gtx)	16.00	dBi
	Tx Cable & Connector Losses	2.00	dB
	Tx Antenna Height (hb)	35.0	meters
Channel / Path	Carrier Frequency (f)	1800.00	MHz
	Link Distance (d)	2.000	km
	Propagation Model	COST-231 Hata Model	Empirical/Analytical
	Path Loss Exponent (n)	3.00	Dimensionless
	Rain Attenuation	0.00	dB
	Atmospheric & Gas Loss	0.00	dB
	Obstacle / Penetration Loss	12.00	dB
Receiver (Rx)	Fade Margin Allocation	10.00	dB
	Rx Antenna Gain (Grx)	0.00	dBi
	Rx Cable & Filter Loss	0.00	dB
	Rx Sensitivity Threshold	-100.00	dBm
	Front-End Noise Figure (NF)	7.00	dB

	Channel Bandwidth (B)	20.00	MHz
	Noise Temperature (T)	290.0	Kelvin (K)

Link Budget Stage-by-Stage Power Accounting

Stage Description	Incremental Gain/Loss (dB)	Cumulative Power Level (dBm)
Tx Power	+43.00 dB	43.00 dBm
After Tx Loss	-2.00 dB	41.00 dBm
EIRP (Tx Antenna)	+16.00 dB	57.00 dBm
After Path Loss	-145.74 dB	-88.74 dBm
After Env Losses	-22.00 dB	-110.74 dBm
Rx Antenna Gain	0.00 dB	-110.74 dBm
Rx Power (Input)	0.00 dB	-110.74 dBm

RF Link Budget Visual Plots



Core Mathematical Formulas Applied:

- Friis Received Power: $Prx\ (dBm) = Ptx + Gtx - Ltx + Grx - Lrx - Lenv$
- Free Space Path Loss: $FSPL\ (dB) = 32.44 + 20 \cdot \log_{10}(d_km) + 20 \cdot \log_{10}(f_MHz)$
- Thermal Noise Floor: $N\ (dBm) = -174\ dBm/Hz + 10 \cdot \log_{10}(B_Hz) + NF\ (dB)$
- Link Margin: $Margin\ (dB) = Prx\ (dBm) - Prx,sens\ (dBm)$
- Shannon Capacity: $C = B \cdot \log_2(1 + SNR_linear)\ (bps)$